

FretGarden Pedagogical Constitution

PEDAGOGICAL STANDARD

Chapter 3 - Lesson Architecture	Version 1.0
Status	Foundational Draft
Authority	Normative

This document forms part of the FretGarden Pedagogical Constitution. Normative terms are used according to the Foundation.

Contents

1. Purpose
2. Constitutional Authority
3. Definition of a Lesson
4. Lesson Versus Explanation, Exercise, Assessment, and Practice Session
 - 4.1 Explanation
 - 4.2 Exercise
 - 4.3 Assessment
 - 4.4 Practice Session
 - 4.5 Unit
5. The Invariant Lesson Spine
6. Lesson Architecture Is Functional, Not Cosmetic
7. Phase I - Position
 - 7.1 Purpose
 - 7.2 Required Elements
 - 7.3 Purpose Statements
 - 7.4 Scope Control
8. Phase II - Activate
 - 8.1 Purpose
 - 8.2 Activation Requirements
 - 8.3 Activation Length
 - 8.4 Activation Failure
9. Phase III - Encounter
 - 9.1 Purpose
 - 9.2 Encounter Forms
 - 9.3 Encounter Before Full Explanation
 - 9.4 Encounter Quality
10. Phase IV - Attend
 - 10.1 Purpose
 - 10.2 Attention Prompts
 - 10.3 Attention Economy
 - 10.4 Irrelevant Attention
11. Phase V - Model
 - 11.1 Purpose
 - 11.2 Minimum Complete Model
 - 11.3 Model Components
 - 11.4 Modeling Versus Over-Explaining
 - 11.5 Demonstration Fidelity
12. Phase VI - Attempt
 - 12.1 Purpose
 - 12.2 Attempt Design
 - 12.3 Early Success
 - 12.4 Productive Failure
13. Phase VII - Interpret

- 13.1 Purpose
- 13.2 Feedback Requirements
- 13.3 Dimensional Feedback
- 13.4 Feedback Timing
- 13.5 Praise
- 14. Phase VIII - Adjust
 - 14.1 Purpose
 - 14.2 Permitted Adjustments
 - 14.3 Adjustment Must Match Error
 - 14.4 Correct Repetition
- 15. Phase IX - Fade
 - 15.1 Purpose
 - 15.2 Types of Scaffolding
 - 15.3 Fading Sequence
 - 15.4 Criteria for Fading
 - 15.5 Restoring Support
 - 15.6 Permanent Reference Tools
- 16. Phase X - Retrieve
 - 16.1 Purpose
 - 16.2 Retrieval Requirements
 - 16.3 Retrieval Level
 - 16.4 Retrieval Difficulty
 - 16.5 Retrieval Feedback
- 17. Phase XI - Vary
 - 17.1 Purpose
 - 17.2 Near-Transfer Variables
 - 17.3 Variation Control
 - 17.4 False Variation
 - 17.5 Interleaving
- 18. Phase XII - Apply
 - 18.1 Purpose
 - 18.2 Application Forms
 - 18.3 Minimum Application Standard
 - 18.4 Deferred Application
 - 18.5 Creative Constraint
- 19. Phase XIII - Consolidate
 - 19.1 Purpose
 - 19.2 Learner Reconstruction
 - 19.3 Practice Prescription
 - 19.4 Garden Philosophy
- 20. Phase XIV - Exit
 - 20.1 Purpose
 - 20.2 Exit Evidence
 - 20.3 Exit Status
 - 20.4 Honest Completion

- 20.5 Next Action
- 21. Phase XV - Return
 - 21.1 Purpose
 - 21.2 Return Types
 - 21.3 Return Conditions
 - 21.4 Return Failure
- 22. Permitted Lesson Sequences
 - 22.1 Experience-First Sequence
 - 22.2 Model-First Sequence
 - 22.3 Contrast Sequence
 - 22.4 Problem-Solution Sequence
 - 22.5 Spiral Sequence
 - 22.6 Diagnostic-Remedial Sequence
- 23. Lesson Length and Boundaries
 - 23.1 No Universal Duration
 - 23.2 Cognitive Boundary
 - 23.3 Objective Boundary
 - 23.4 Multi-Session Lessons
 - 23.5 Micro-Lessons
- 24. Managing Cognitive Load
 - 24.1 Intrinsic and Avoidable Demand
 - 24.2 Novelty Budget
 - 24.3 Chunking
 - 24.4 Worked Examples
 - 24.5 Redundancy
- 25. Scaffolding Standard
 - 25.1 Definition
 - 25.2 Scaffold Selection
 - 25.3 Minimal Necessary Support
 - 25.4 Scaffold Visibility
 - 25.5 Scaffold Fading Plan
 - 25.6 Scaffold Dependency Test
- 26. Translation of Objectives Into Instruction
- 27. Lesson Content Selection
 - 27.1 Essential Content
 - 27.2 Supporting Content
 - 27.3 Deferred Content
 - 27.4 False Completeness
- 28. Examples and Non-Examples
 - 28.1 Example Function
 - 28.2 Non-Examples
 - 28.3 Example Progression
 - 28.4 Genre Use
- 29. Questions and Prompts
 - 29.1 Prompt Function

- 29.2 Leading Questions
- 29.3 Decorative Questions
- 29.4 Wait and Attempt
- 30. Practice Within a Lesson
 - 30.1 Deliberate Practice Structure
 - 30.2 Repetition Counts
 - 30.3 Tempo
 - 30.4 Focused Intervals
 - 30.5 Physical Safety
- 31. Feedback Architecture
 - 31.1 Feedback Layers
 - 31.2 Priority
 - 31.3 Overcorrection
 - 31.4 Answer Revelation
- 32. Error-Recovery Loop
- 33. Retrieval Architecture
 - 33.1 Immediate Retrieval
 - 33.2 Expanding Retrieval
 - 33.3 Retrieval Spacing Within the Lesson
 - 33.4 Mixed Retrieval
 - 33.5 Failed Retrieval
- 34. Application Architecture
 - 34.1 Application Ladder
 - 34.2 Authenticity
 - 34.3 Explanation After Application
 - 34.4 Application Failure
- 35. Delayed-Review Architecture
 - 35.1 Review Is Part of Lesson Design
 - 35.2 Review Specification
 - 35.3 Review Progression
 - 35.4 Over-Review
- 36. Lesson Exit and Progression
 - 36.1 Exit Is Not Mastery
 - 36.2 Readiness Claims
 - 36.3 Optional Advancement
 - 36.4 Remediation Routing
- 37. Domain-Specific Architectural Variations
 - 37.1 Perception Lessons
 - 37.2 Retrieval Lessons
 - 37.3 Conceptual Lessons
 - 37.4 Fretboard Lessons
 - 37.5 Physical-Technique Lessons
 - 37.6 Rhythm Lessons
 - 37.7 Creative Lessons
 - 37.8 Metacognitive Lessons

- 38. Lesson Sequences and Dependency
 - 38.1 Backward Dependency
 - 38.2 Forward Dependency
 - 38.3 Circular Dependency Prohibition
 - 38.4 Shared Prerequisites
- 39. Tone and Instructional Voice Within Lessons
- 40. The Garden Philosophy in Lesson Architecture
- 41. Common Architectural Failures
 - 41.1 Information Dump
 - 41.2 Activity Chain Without Objective
 - 41.3 Demonstration Dependency
 - 41.4 Quiz as Instruction
 - 41.5 Premature Integration
 - 41.6 Permanent Isolation
 - 41.7 False Transfer
 - 41.8 Completion Inflation
 - 41.9 Review Omission
 - 41.10 Metaphor Saturation
 - 41.11 Excessive Correction
 - 41.12 Fixed Attempt Logic
- 42. Lesson Authoring Record
- 43. Lesson Quality Review
- 44. Lesson Conformance Tests
- 45. Examples of Compliant Lesson Architecture
 - 45.1 Natural-Note Retrieval Lesson
 - 45.2 Major-Triad Construction Lesson
 - 45.3 Eighth-Note Subdivision Lesson
- 46. Examples of Noncompliant Lesson Architecture
 - 46.1 Definition-Only Lesson
 - 46.2 Diagram-Dependency Lesson
 - 46.3 Speed Challenge Lesson
 - 46.4 Musical Example Without Learner Use
 - 46.5 Fixed Completion Lesson
- 47. Constitutional Non-Negotiables
- 48. Tradeoffs and Resolution Rules
 - 48.1 Consistency Versus Variety
 - 48.2 Guidance Versus Discovery
 - 48.3 Explanation Versus Action
 - 48.4 Support Versus Independence
 - 48.5 Lesson Momentum Versus Remediation
 - 48.6 Authenticity Versus Diagnosability
 - 48.7 Concision Versus Completeness
- 49. Assumptions Made
- 50. Unresolved Questions
- 51. Topics Intentionally Deferred

- 52. Required Future Cross-References
 - 52.1 Exercise Design Chapter
 - 52.2 Assessment Chapter
 - 52.3 Mastery and Progression Chapter
 - 52.4 Review and Retention Chapter
 - 52.5 Feedback and Remediation Chapters
 - 52.6 Domain-Specific Pedagogy Chapters
- 53. Chapter Conformance Statement

Chapter 3 - Lesson Architecture

Version: 1.0

Status: Foundational Draft

Document Type: Normative Pedagogical Standard

Authority: Binding on every FretGarden lesson and lesson sequence

Primary Dependencies: Foundation Sections 2, 4, 6, 7, 8, 9, 10, and 13; Chapter 2 Sections 3-6, 17-19, 40-41, and 49

1. Purpose

This chapter defines the required architecture of a FretGarden lesson.

It establishes how a learning objective is translated into instruction, learner action, feedback, retrieval, application, and delayed review. It defines the functions that every core lesson must perform, the order constraints that govern those functions, the permissible forms of variation, and the standards by which a lesson is considered complete.

A lesson is not merely a container for information. It is a designed sequence of experiences intended to produce evidence of a defined change in capability. Its architecture must therefore connect the objective to the learner's attention, actions, errors, corrections, retrieval attempts, and future use.

The purpose of this chapter is not to force every musical concept into an identical script. Ear training, fretboard navigation, physical technique, rhythmic development, conceptual explanation, and creative work require different instructional forms. Nevertheless, every core lesson has the same constitutional obligations. It must orient the learner, activate relevant prior knowledge, create meaningful contact with the target, direct attention, support an initial attempt, provide interpretable feedback, reduce support, require retrieval, connect to musical use, and establish later review.

This chapter distinguishes these invariant obligations from optional lesson patterns.

2. Constitutional Authority

This chapter operationalizes the following requirements:

- every lesson must produce an observable change;
- objectives and evidence must be designed together;
- sound, action, retrieval, delayed review, and application are required for core content;
- scaffolding must be temporary;
- one principal learning burden should be introduced at a time;
- decomposition must lead back to integration;
- practice instructions are part of the lesson;
- feedback must change the next attempt;
- and lesson completion must not be confused with learning.

When a lesson template, author preference, content-management constraint, or engagement mechanism conflicts with this chapter, this chapter governs.

3. Definition of a Lesson

A FretGarden lesson is a bounded instructional sequence intended to move one or more defined objectives from a specified entry state toward a specified target state.

A lesson **MUST** include:

- 1 a primary objective;
- 2 explicit prerequisites;
- 3 an evidence statement;
- 4 a defined learner entry condition;
- 5 meaningful encounter with the target;
- 6 active learner response;
- 7 feedback and correction;
- 8 reduction of support;
- 9 retrieval or reconstruction;
- 10 musical application or a defined path to application;
- 11 a completion decision based on evidence; and
- 12 a delayed-review obligation.

A lesson **MAY** span several sessions. A session **MAY** contain part of a lesson, one complete lesson, or several short lessons. Lesson boundaries are educational, not merely temporal or interface-based.

4. Lesson Versus Explanation, Exercise, Assessment, and Practice Session

4.1 Explanation

An explanation organizes or clarifies a concept. It is one possible component of a lesson. An explanation without learner action and evidence is not a complete core lesson.

4.2 Exercise

An exercise is a bounded activity serving instruction, practice, diagnosis, or assessment. A lesson may contain several exercises. An exercise is not automatically a lesson because it may omit orientation, conceptual framing, transfer, or review planning.

4.3 Assessment

An assessment collects evidence. A lesson may include formative assessment, but an assessment by itself does not necessarily teach or remediate.

4.4 Practice Session

A practice session organizes time and effort. It may revisit several lessons or objectives. A practice session is not defined by introducing new content.

4.5 Unit

A unit is a larger sequence of related lessons. Unit architecture will be governed by a later curriculum-sequencing chapter.

5. The Invariant Lesson Spine

Every core FretGarden lesson MUST address the following functions:

- 1 Position - establish purpose, scope, and readiness;
- 2 Activate - retrieve prerequisites and relevant prior knowledge;
- 3 Encounter - create meaningful contact with the target phenomenon;
- 4 Attend - direct attention to the relationship that matters;
- 5 Model - provide the minimum complete representation or demonstration;
- 6 Attempt - require the learner to act;
- 7 Interpret - provide feedback and identify the meaning of the attempt;
- 8 Adjust - correct, simplify, or refine the next attempt;
- 9 Fade - reduce unnecessary support;
- 10 Retrieve - require production without the complete answer visible;
- 11 Vary - change context enough to test stability and near transfer;
- 12 Apply - connect the capability to musical use;
- 13 Consolidate - summarize the working model and practice method;
- 14 Exit - determine what has and has not been demonstrated;
- 15 Return - schedule or define delayed review.

These functions form the invariant lesson spine. They do not require fifteen visibly separate screens, paragraphs, or activities. Several functions may occur in one well-designed task. A lesson MAY revisit functions recursively. The order may vary where the variation is pedagogically justified.

However, no function may be omitted from a core lesson merely because another lesson commonly includes it. If application or delayed review occurs in a later lesson, the current lesson MUST identify that dependency explicitly.

6. Lesson Architecture Is Functional, Not Cosmetic

Compliance is determined by educational function rather than by headings or labels.

A lesson does not satisfy encounter merely because it plays an audio clip. The learner must contact the target phenomenon in a way that supports the objective.

A lesson does not satisfy retrieval merely because it includes a quiz. The task must require memory or reconstruction at the stated support level.

A lesson does not satisfy application merely because an example sounds musical. The learner must use the target capability within meaningful musical action or reasoning.

A lesson does not satisfy consolidation merely because it displays a summary. The learner must leave with a coherent model of what was learned, how it was practiced, and what remains developing.

7. Phase I - Position

7.1 Purpose

Position establishes why the lesson exists, what capability is being developed, what is included, and what is not yet required.

Position prevents learners from treating every detail as equally important and prevents lesson authors from drifting beyond the objective.

7.2 Required Elements

A lesson **MUST** establish, in learner-appropriate form:

- the musical or practical purpose;
- the primary capability;
- the approximate scope;
- any necessary readiness condition;
- and the standard by which the learner should judge the current attempt.

The lesson need not expose formal specification language. It must communicate enough for deliberate participation.

7.3 Purpose Statements

Purpose statements should connect the lesson to musicianship.

Acceptable:

Chord tones help you choose notes that remain connected to the harmony when the chord changes.

Weak:

In this lesson you will learn chord tones.

The second statement names a topic but does not explain why the capability matters.

7.4 Scope Control

Position **SHOULD** identify boundaries that reduce unnecessary pressure.

Example:

Today the goal is to locate the root and fifth on two string sets. Fast recall across the entire fretboard is not required yet.

This language protects accuracy and attention by separating the current target from future fluency.

8. Phase II - Activate

8.1 Purpose

Activation retrieves prerequisite knowledge and makes relevant relationships available before the new burden is introduced.

Activation is not a decorative review. It determines readiness and reduces the working-memory cost of the new lesson.

8.2 Activation Requirements

A lesson **MUST** activate prerequisites that are required for the primary objective unless valid evidence shows they are already available and the lesson can access that evidence.

Activation **MAY** include:

- naming known notes;
- locating a root;
- clapping a familiar subdivision;
- recalling an interval formula;
- identifying chord tones learned previously;
- or demonstrating a basic movement.

8.3 Activation Length

Activation **SHOULD** be brief when prerequisites are stable. It **MAY** expand into remediation when instability would compromise the lesson.

8.4 Activation Failure

When a required prerequisite is unavailable, the lesson **MUST** do one of the following:

- 1 provide immediate micro-remediation;
- 2 reduce the lesson's demands;
- 3 redirect to targeted prerequisite work; or
- 4 explicitly reclassify the activity as exploratory exposure rather than readiness-based instruction.

The lesson **MUST NOT** continue unchanged while treating predictable prerequisite failure as failure of the new objective.

9. Phase III - Encounter

9.1 Purpose

Encounter creates meaningful contact with the musical phenomenon before or alongside abstraction.

The learner should hear, play, feel, compare, observe, or otherwise experience the target relationship.

9.2 Encounter Forms

Encounter may include:

- hearing two contrasting intervals;
- playing a root followed by a chord tone;
- feeling quarter-note pulse before naming subdivision;
- observing the same note in several fretboard positions;
- comparing major and minor triads;

- or performing a brief musical phrase containing the target.

9.3 Encounter Before Full Explanation

Whenever reasonably possible, the learner SHOULD encounter the target before receiving an extensive explanation.

Concise orientation may precede encounter. Safety, tuning, physical setup, or task instructions may also require prior explanation. The constitutional prohibition is against allowing abstraction to replace experience.

9.4 Encounter Quality

The encounter must make the target perceptible. Examples should minimize unrelated novelty and foreground the relationship that later explanation will name.

10. Phase IV - Attend

10.1 Purpose

Encounter alone does not guarantee learning. Attend directs the learner's attention toward the feature that matters.

A learner can play a scale without noticing the half-step locations, hear a cadence without noticing tonal resolution, or follow a metronome without understanding subdivision.

10.2 Attention Prompts

Attention may be directed through questions, comparison, prediction, repetition, isolation, exaggeration, or physical marking.

Examples:

- Which note feels most settled?
- Does the second example sound wider or narrower?
- Count where the note occurs between the clicks.
- Find the root before looking for the third.
- Notice whether your fretting hand tightens before the shift.

10.3 Attention Economy

A lesson SHOULD direct attention to one principal feature at a time during initial learning.

It MAY later integrate several attentional demands after each is sufficiently stable.

10.4 Irrelevant Attention

Visual decoration, terminology, narrative, or feedback that competes with the target SHOULD be reduced.

11. Phase V - Model

11.1 Purpose

Model provides the learner with the minimum complete representation needed to make a meaningful attempt.

The model may be an explanation, demonstration, diagram, sound example, worked derivation, count, or physical procedure.

11.2 Minimum Complete Model

A model must be simple enough for the learner's current stage but complete enough not to establish a misconception.

For example, an early triad model may focus on root, third, and fifth without discussing every inversion. It must not imply that the lowest note is always the root.

11.3 Model Components

A model SHOULD make clear:

- the target;
- the relevant relationship;
- the procedure or strategy;
- the quality criterion;
- and a common error or boundary where useful.

11.4 Modeling Versus Over-Explaining

A model MUST NOT front-load every exception, alternate terminology, or advanced implication when those details do not improve the current attempt.

Additional complexity should be layered later.

11.5 Demonstration Fidelity

Demonstrations MUST exhibit the qualities the learner is expected to reproduce. A timing lesson must model stable time. A technique lesson must not demonstrate avoidable tension. An explanatory model must use the constitution's terminology consistently.

12. Phase VI - Attempt

12.1 Purpose

Attempt requires the learner to produce evidence rather than remain a spectator.

The first attempt should be difficult enough to require attention but sufficiently supported to produce interpretable information.

12.2 Attempt Design

An initial attempt SHOULD:

- target the primary objective;

- control unrelated demands;
- define the permitted support;
- make error visible;
- and permit a useful next action.

12.3 Early Success

Early success is useful when it confirms the learner understands the task. It must not be manufactured by making the answer effectively visible when retrieval or reasoning is the objective.

12.4 Productive Failure

An incorrect initial attempt may be productive if it is interpretable and followed by correction. Repeated failure without adjustment is noncompliant.

13. Phase VII - Interpret

13.1 Purpose

Interpret converts the attempt into information.

The learner must understand not only whether an attempt succeeded, but which dimension was successful or unstable and what that means for the next attempt.

13.2 Feedback Requirements

Feedback SHOULD identify:

- what was correct;
- what differed from the target;
- the likely error category;
- and the next action.

13.3 Dimensional Feedback

When a task includes several dimensions, feedback SHOULD preserve them.

Example:

The notes were correct, but the second note arrived before the beat. Keep the fingering and reduce the tempo so the shift remains aligned.

This is superior to a single incorrect judgment because it preserves the learner's successful note knowledge while directing temporal correction.

13.4 Feedback Timing

Immediate feedback is appropriate when an error would otherwise be repeated or when the learner cannot diagnose it. Delayed feedback may be appropriate when it supports self-monitoring, performance continuity, or independent evaluation.

The timing must serve the objective.

13.5 Praise

Praise should refer to observable effort, strategy, correction, attention, or improved performance. Generic celebration MUST NOT replace instructional feedback.

14. Phase VIII - Adjust

14.1 Purpose

Adjust changes the conditions of the next attempt based on evidence.

Adjustment turns error into practice rather than repeated failure.

14.2 Permitted Adjustments

Adjustment may include:

- reducing tempo;
- reducing the number of notes;
- isolating a transition;
- restoring a cue;
- changing the representation;
- counting aloud;
- returning to a reference pitch;
- using a simpler rhythm;
- narrowing the fretboard region;
- or correcting a conceptual model.

14.3 Adjustment Must Match Error

A learner who misunderstands a chord formula does not primarily need a slower tempo. A learner who knows the notes but cannot synchronize a shift may not need another theoretical explanation.

The adjustment must address the diagnosed cause.

14.4 Correct Repetition

After adjustment, the learner SHOULD perform enough correct or improved repetitions to establish the corrected pattern without encouraging mindless volume.

15. Phase IX - Fade

15.1 Purpose

Fade reduces temporary support so the learner assumes more of the task.

Scaffolding is successful only when it can be removed without collapse.

15.2 Types of Scaffolding

Scaffolding may include:

- complete demonstrations;

- highlighted notes;
- root markers;
- diagrams;
- verbal steps;
- count syllables;
- restricted choices;
- reduced tempo;
- fixed starting positions;
- or immediate corrective prompts.

15.3 Fading Sequence

Support SHOULD be reduced from the least instructionally essential element first.

A common sequence is:

- 1 complete model;
- 2 model with learner participation;
- 3 partial visual or verbal cue;
- 4 starting anchor only;
- 5 task instruction without solution cue;
- 6 independent initiation and completion.

This sequence may vary by domain.

15.4 Criteria for Fading

Support should be reduced when the learner demonstrates sufficient accuracy and task understanding under the current support level.

Support MUST NOT be removed merely because a fixed number of attempts has occurred.

15.5 Restoring Support

Support MAY be temporarily restored when performance deteriorates. Restoration should be targeted and followed by another fading attempt.

15.6 Permanent Reference Tools

Some supports, such as standard notation, metronomes, tuners, charts, or written music, may be legitimate tools rather than scaffolds. The lesson must distinguish between a representation the musician is expected to use and an answer cue that should eventually be removed.

16. Phase X - Retrieve

16.1 Purpose

Retrieve requires the learner to access or reconstruct the target without the complete answer visible.

Retrieval converts guided familiarity into more durable access.

16.2 Retrieval Requirements

Every core lesson MUST include same-lesson retrieval or define an immediate retrieval dependency in a directly connected lesson segment.

Retrieval may involve:

- naming;
- locating;
- constructing;
- explaining;
- playing;
- singing;
- counting;
- predicting;
- or selecting a strategy.

16.3 Retrieval Level

The retrieval task must match the objective. Recognition is insufficient when recall is required. A full fretboard diagram is excessive support when independent navigation is required.

16.4 Retrieval Difficulty

The first retrieval should usually preserve the target relationship while removing answer support. Additional difficulty should be introduced only after the basic retrieval demand is clear.

16.5 Retrieval Feedback

Feedback after retrieval should protect the attempt from turning into repeated guessing. After an error, the learner should receive enough information to reconstruct the model and attempt again.

17. Phase XI - Vary

17.1 Purpose

Vary changes incidental features to determine whether the learner's capability is tied to one example.

17.2 Near-Transfer Variables

Variation may include:

- a new root;
- another string set;
- a changed fretboard position;
- reduced highlighting;
- a different rhythmic order;
- another register;
- a new chord within the same quality;

- or a different starting point.

17.3 Variation Control

During early transfer, one meaningful variable SHOULD change at a time when practical. This keeps failure diagnosable.

17.4 False Variation

Changing cosmetic details, item order, colors, or wording without changing the musical or cognitive demand does not establish transfer.

17.5 Interleaving

Related concepts MAY be mixed after initial clarity. Interleaving should require discrimination, not merely create confusion.

18. Phase XII - Apply

18.1 Purpose

Apply connects the target capability to music-making, analysis, creative choice, or effective practice.

Application answers why the lesson matters.

18.2 Application Forms

Application may include:

- a riff or melody;
- a chord progression;
- improvisation;
- composition;
- variation;
- analysis of unfamiliar music;
- accompaniment;
- transcription;
- or planning a practice strategy.

18.3 Minimum Application Standard

The learner must use the target relationship rather than merely encounter it inside a musical example.

Listening to a riff containing a perfect fifth is context. Choosing, locating, or explaining the fifth within the riff is application.

18.4 Deferred Application

Some foundational capabilities may require several lessons before meaningful full application is possible. In such cases, the lesson MUST provide a constrained use or explicitly identify the later lesson where integration occurs.

Permanent deferral is prohibited.

18.5 Creative Constraint

Application tasks SHOULD provide enough constraint to foreground the objective while leaving room for meaningful learner choice.

19. Phase XIII - Consolidate

19.1 Purpose

Consolidate organizes the lesson into a usable mental and practice model.

It answers:

- What relationship was learned?
- How can it be recognized or produced?
- What strategy was effective?
- What errors require attention?
- Where will this be used next?

19.2 Learner Reconstruction

Whenever practical, consolidation SHOULD require the learner to summarize, demonstrate, classify, or reconstruct rather than passively read a summary.

19.3 Practice Prescription

When continued practice is required, consolidation MUST specify:

- the practice target;
- a manageable scope;
- the support level;
- tempo or timing conditions where relevant;
- error response;
- success condition;
- and stopping or break guidance.

19.4 Garden Philosophy

Garden language MAY frame consolidation by distinguishing planting, strengthening, watering, pruning, connecting, or revisiting. Metaphor MUST remain subordinate to precise instruction.

20. Phase XIV - Exit

20.1 Purpose

Exit determines what evidence the lesson produced and what claim may legitimately be made.

20.2 Exit Evidence

A core lesson SHOULD end with an independent or minimally supported attempt aligned to the primary objective.

The exit task must not be identical to a fully modeled example when the objective requires reconstruction or transfer.

20.3 Exit Status

Possible lesson-exit conclusions include:

- introduced;
- guided capability demonstrated;
- independent in familiar conditions;
- developing but unstable;
- prerequisite remediation needed;
- or insufficient evidence.

A lesson exit **MUST NOT** claim retention, broad transfer, fluency, or mastery unless the evidence required by Chapter 2 already exists.

20.4 Honest Completion

A learner may complete the instructional experience without meeting the objective. The lesson should record completion and capability separately.

20.5 Next Action

The exit must identify the next educational action: advance, practice, review, remediate, collect more evidence, or apply in a connected lesson.

21. Phase XV - Return

21.1 Purpose

Return establishes when and how the capability will be retrieved after delay.

A lesson is constitutionally incomplete as a plan for core content if no delayed return exists.

21.2 Return Types

Return may occur through:

- scheduled retrieval;
- cumulative review;
- interleaved practice;
- application in a later lesson;
- prerequisite revalidation;
- or a dedicated retention assessment.

21.3 Return Conditions

The delayed task **SHOULD** reduce complete re-teaching, preserve the target capability, and include enough variation to distinguish memory from exact item familiarity.

21.4 Return Failure

Failure during delayed review should trigger reactivation and targeted practice, not erase all prior evidence or frame forgetting as abnormal.

22. Permitted Lesson Sequences

The invariant spine does not mandate one visible order. The following patterns are constitutionally permitted when they satisfy all relevant functions.

22.1 Experience-First Sequence

Encounter -> Attend -> Attempt -> Model -> Adjust -> Retrieve -> Vary -> Apply

Useful when learners can discover or perceive a relationship before formal explanation.

22.2 Model-First Sequence

Position -> Activate -> Model -> Attempt -> Interpret -> Adjust -> Fade -> Retrieve -> Apply

Useful when safety, procedural clarity, or unfamiliar notation requires prior modeling.

22.3 Contrast Sequence

Encounter A -> Encounter B -> Attend to difference -> Classify -> Explain -> Retrieve -> Apply

Useful for intervals, chord qualities, articulations, and rhythmic distinctions.

22.4 Problem-Solution Sequence

Attempt meaningful task -> Identify failure -> Isolate prerequisite -> Model -> Practice -> Reintegrate

Useful when the learner benefits from first experiencing why a capability is needed.

22.5 Spiral Sequence

Brief encounter -> partial model -> attempt -> additional model -> expanded attempt -> application

Useful for concepts that require several layers of explanation.

22.6 Diagnostic-Remedial Sequence

Activate -> diagnostic attempt -> classify error -> targeted micro-lesson -> corrected attempt -> fade -> recheck

Useful for prerequisite repair.

No sequence is valid merely because it matches one of these names. It must remain aligned to objective and evidence.

23. Lesson Length and Boundaries

23.1 No Universal Duration

A lesson is not defined by a fixed number of minutes, words, examples, or activities.

23.2 Cognitive Boundary

A lesson SHOULD end or pause when the learner's ability to attend, correct, or retrieve deteriorates substantially.

23.3 Objective Boundary

A lesson SHOULD have one primary objective. Closely related secondary objectives are permitted when they support the primary objective and do not create competing novelty.

23.4 Multi-Session Lessons

A complex lesson MAY span sessions. Each session must have a clear stopping point, preserve the learner's place in the learning cycle, and begin with reactivation when necessary.

23.5 Micro-Lessons

A micro-lesson may address one misconception, term, movement, or retrieval gap. It remains subject to objective-evidence alignment, but its encounter, model, practice, and exit may be compressed.

24. Managing Cognitive Load

24.1 Intrinsic and Avoidable Demand

Some difficulty belongs to the musical objective. Other difficulty arises from unclear wording, inconsistent diagrams, excessive terminology, unfamiliar response formats, or too many simultaneous demands.

Avoidable demand SHOULD be reduced.

24.2 Novelty Budget

A lesson SHOULD introduce one principal novelty at a time during initial learning.

If the new target is a chord-tone relationship, the rhythm, fingering demand, notation, and fretboard region should usually be familiar or controlled.

24.3 Chunking

Complex capabilities should be divided into meaningful components that can later be recombined.

Chunks must preserve musical relationships. Arbitrary fragmentation can make reintegration harder.

24.4 Worked Examples

Worked examples MAY reduce load during early learning. They must transition into completion problems, reconstruction, or independent attempts.

24.5 Redundancy

Multiple representations are useful when they converge. Redundant explanation that adds no new function may compete with attention and SHOULD be removed.

25. Scaffolding Standard

25.1 Definition

Scaffolding is temporary support that permits a learner to perform or understand a task not yet available independently.

25.2 Scaffold Selection

A scaffold must address a specific obstacle. It should not reduce every difficulty indiscriminately.

25.3 Minimal Necessary Support

The lesson SHOULD provide the least support that allows a meaningful attempt.

Too little support creates noise and repeated failure. Too much support conceals whether learning occurred.

25.4 Scaffold Visibility

Learners SHOULD know when they are using support and what independence will eventually require.

25.5 Scaffold Fading Plan

Every scaffold used for core learning MUST have a planned fading condition or be identified as a permanent musical tool.

25.6 Scaffold Dependency Test

If performance collapses whenever one cue is removed, the lesson must determine whether the cue represents an unstable prerequisite, an incomplete model, or insufficient practice.

26. Translation of Objectives Into Instruction

Every objective should be translated through the following design questions:

- 1 What must the learner experience before the concept has meaning?
- 2 What prior capability must be active?
- 3 What feature must receive attention?
- 4 What is the minimum complete model?
- 5 What first action will produce interpretable evidence?
- 6 What errors are likely?
- 7 What feedback will change the next attempt?
- 8 What support is necessary, and how will it fade?
- 9 What retrieval task matches the objective?
- 10 What variation will test stability?
- 11 What musical use will make the capability meaningful?
- 12 What evidence is sufficient for lesson exit?
- 13 When will delayed review occur?

A lesson author MUST answer these questions before final approval.

27. Lesson Content Selection

27.1 Essential Content

Content is essential when it is required to understand, perform, retrieve, or apply the objective.

27.2 Supporting Content

Supporting content provides context, analogy, history, terminology, or enrichment. It must not obscure the essential model.

27.3 Deferred Content

Advanced nuance, exceptions, alternate systems, and distant applications SHOULD be deferred when they do not improve the current objective.

27.4 False Completeness

A lesson MUST NOT pursue encyclopedic completeness at the cost of learnability. It should provide the smallest accurate model that supports current and future development.

28. Examples and Non-Examples

28.1 Example Function

Every example MUST have a stated authoring purpose, such as isolating a relationship, providing contrast, demonstrating use, revealing an error, or supporting transfer.

28.2 Non-Examples

A lesson SHOULD include non-examples when they clarify boundaries or common confusions.

Examples include a note that is not a chord tone, a rhythm that begins before rather than on the beat, or a shape with the same geometry but a different function.

28.3 Example Progression

Examples SHOULD progress from clear and constrained to varied and musically integrated.

28.4 Genre Use

Genre-specific examples MAY improve relevance, but the chosen style must not introduce unnecessary prerequisites or imply that the underlying relationship belongs only to that genre.

29. Questions and Prompts

29.1 Prompt Function

Questions should direct attention, elicit retrieval, expose reasoning, support prediction, or encourage self-diagnosis.

29.2 Leading Questions

Leading questions MAY support early learning but SHOULD be reduced when independent reasoning is required.

29.3 Decorative Questions

Questions whose answers are immediately supplied or whose purpose is merely conversational SHOULD NOT be mistaken for active learning.

29.4 Wait and Attempt

The lesson should permit a genuine attempt before revealing the answer when retrieval, prediction, or reasoning is intended.

30. Practice Within a Lesson

30.1 Deliberate Practice Structure

Practice activities **MUST** define target, scope, support, quality criterion, error response, and stopping or transition condition.

30.2 Repetition Counts

Fixed repetition counts **MAY** organize effort but **MUST NOT** independently determine readiness. Repetitions should stop, simplify, or change when quality deteriorates or the objective has been sufficiently demonstrated.

30.3 Tempo

Tempo should be selected to preserve the objective's quality requirements. Tempo increases must follow stability rather than attempt to create it.

30.4 Focused Intervals

Lessons **MAY** recommend focused work intervals such as twenty-five minutes followed by five minutes of rest or shorter structures such as ten minutes followed by five. The key standard is sustainable attention, not one mandatory duration.

30.5 Physical Safety

Lessons involving physical repetition **SHOULD** remind learners to stop or modify work when pain, persistent strain, or loss of control appears. FretGarden does not diagnose injury.

31. Feedback Architecture

31.1 Feedback Layers

Feedback may operate at four layers:

- 1 Outcome - whether the target was met;
- 2 Process - how the learner approached the task;
- 3 Strategy - what adjustment should be made;
- 4 Self-regulation - how the learner can evaluate future attempts.

31.2 Priority

Early feedback may emphasize outcome and process. Later feedback should increasingly develop strategy and self-regulation.

31.3 Overcorrection

Feedback **SHOULD** prioritize errors relevant to the objective. Correcting every minor issue simultaneously can obscure the primary target.

31.4 Answer Revelation

When an answer is revealed, the learner SHOULD be required to reconstruct, repeat, explain, or apply it rather than passively continue.

32. Error-Recovery Loop

Every skill lesson SHOULD support the following recovery loop:

- 1 stop or stabilize;
- 2 identify the dimension that failed;
- 3 determine the likely cause;
- 4 reduce or change the relevant difficulty;
- 5 perform a corrected version;
- 6 repeat attentively;
- 7 restore difficulty gradually; and
- 8 verify the correction under reduced support.

Repeated blind restarting is not an acceptable recovery method.

33. Retrieval Architecture

33.1 Immediate Retrieval

Core content must be retrieved within the lesson after sufficient modeling and practice.

33.2 Expanding Retrieval

Retrieval should progress from close and supported to varied and independent.

33.3 Retrieval Spacing Within the Lesson

Where practical, intervening activity should occur between initial learning and later same-lesson retrieval. This reduces reliance on immediate echoic or working memory.

33.4 Mixed Retrieval

Previously learned concepts MAY be interleaved with the new target after the target has been made clear.

33.5 Failed Retrieval

A failed retrieval attempt should trigger brief reconstruction, not complete restarting of the entire lesson unless the underlying model is absent.

34. Application Architecture

34.1 Application Ladder

Application SHOULD develop through:

- 1 constrained use;

- 2 guided variation;
- 3 independent selection within limits;
- 4 integration with other capabilities;
- 5 and open-ended musical choice where appropriate.

34.2 Authenticity

Application should preserve enough musical context to make the relationship meaningful without adding unrelated difficulty.

34.3 Explanation After Application

Learners SHOULD sometimes identify or explain the choices they made. This reveals whether application was intentional or accidental.

34.4 Application Failure

Failure in application does not necessarily invalidate isolated capability. The lesson should diagnose whether the problem is transfer, timing, execution, perception, or an unstable prerequisite.

35. Delayed-Review Architecture

35.1 Review Is Part of Lesson Design

The original lesson author is responsible for defining the review obligation, even when another system or later lesson delivers it.

35.2 Review Specification

A review specification SHOULD identify:

- the objective;
- the intended delay class;
- the support level;
- the representation;
- the variation;
- the evidence collected;
- and the response to failure.

35.3 Review Progression

Reviews may develop from reactivation to retention verification, fluency building, transfer, and integration.

35.4 Over-Review

Review should not repeat already fluent material so frequently that it crowds out developing capabilities. Frequency should reflect importance, stability, and evidence of forgetting.

36. Lesson Exit and Progression

36.1 Exit Is Not Mastery

A lesson can establish guided, independent, or stable same-session capability. Retention and broader transfer usually require later evidence.

36.2 Readiness Claims

A lesson MAY establish functional readiness for the next lesson when the prerequisite evidence requirement has been met.

36.3 Optional Advancement

Where strict readiness is not required, the learner MAY continue while the capability remains marked as developing. Later lessons must not silently assume mastery.

36.4 Remediation Routing

When exit evidence reveals a specific gap, the next action should target that gap rather than require indiscriminate repetition of the entire lesson.

37. Domain-Specific Architectural Variations

37.1 Perception Lessons

Perception lessons should emphasize encounter, contrast, directed attention, prediction, classification, reproduction, varied timbre or register, and delayed discrimination.

Visual information must not reveal answers in ear-based assessments.

37.2 Retrieval Lessons

Retrieval lessons should emphasize cue control, constructed response, response strategy, expanding variation, delay, and low guessing probability.

37.3 Conceptual Lessons

Conceptual lessons should connect experience to terminology, model relationships, require derivation or explanation, use examples and non-examples, and transfer principles to unfamiliar material.

37.4 Fretboard Lessons

Fretboard lessons should establish reference anchors, relational movement, note or interval identity, multiple starting points, diagram fading, position connection, and musical use.

37.5 Physical-Technique Lessons

Technique lessons should establish physical target, safe and controlled model, isolated movement, slow attempt, sensory attention, error correction, gradual coordination, tempo only after stability, and reintegration into music.

37.6 Rhythm Lessons

Rhythm lessons should establish pulse, bodily or vocal participation, subdivision, decoding, synchronization, silent-beat or reduced-click practice, and musical application.

37.7 Creative Lessons

Creative lessons should establish constraints, demonstrate possibilities without prescribing one answer, require learner choice, provide criteria, support revision, and invite explanation or reflection.

37.8 Metacognitive Lessons

Practice lessons should begin with a real or representative practice problem, model diagnosis, require strategy selection, compare outcomes, and fade instructor control.

38. Lesson Sequences and Dependency

A lesson may depend on earlier or later lessons. Dependencies **MUST** be explicit.

38.1 Backward Dependency

The lesson identifies prerequisites and evidence required from prior instruction.

38.2 Forward Dependency

The lesson identifies where application, review, integration, or advanced nuance will occur later.

38.3 Circular Dependency Prohibition

Two lessons **MUST NOT** each assume independent mastery that only the other lesson establishes.

38.4 Shared Prerequisites

When several lessons depend on the same capability, the curriculum should centralize its definition and review expectations rather than redefine it inconsistently.

39. Tone and Instructional Voice Within Lessons

Lesson language must be calm, precise, respectful, and direct.

It **SHOULD**:

- acknowledge genuine difficulty;
- normalize productive error;
- explain why a strategy is being used;
- distinguish current goals from later fluency;
- and provide actionable next steps.

It **MUST NOT**:

- shame slowness or error;
- exaggerate progress;
- manufacture urgency;
- infantilize beginners;
- use the garden metaphor in place of technical language;
- or celebrate navigation and completion as though they were learning.

40. The Garden Philosophy in Lesson Architecture

Garden language may clarify the function of a lesson:

- planting for initial encounter;
- rooting for prerequisites and foundational relationships;
- watering for deliberate repetition;
- pruning for correction and removal of ineffective habits;
- branching for variation and transfer;
- harvesting for musical application;
- and returning through seasons for delayed review.

These terms MAY guide authoring and occasional learner-facing framing. They MUST NOT create a parallel vocabulary that learners must decode before understanding the music.

41. Common Architectural Failures

41.1 Information Dump

The lesson explains extensively but provides little learner action.

41.2 Activity Chain Without Objective

The lesson contains several engaging tasks that do not build one defined capability.

41.3 Demonstration Dependency

The learner performs only by copying a continuously visible model.

41.4 Quiz as Instruction

The lesson presents unexplained questions and treats repeated guessing as learning.

41.5 Premature Integration

The learner must manage several novel demands before component relationships are clear.

41.6 Permanent Isolation

The lesson drills a component without connecting it to music.

41.7 False Transfer

The lesson changes cosmetic details and claims generalization.

41.8 Completion Inflation

Finishing the lesson produces a mastery or fluency claim.

41.9 Review Omission

The lesson ends with no planned delayed return.

41.10 Metaphor Saturation

Garden language obscures the actual musical concept.

41.11 Excessive Correction

Feedback attempts to repair every detail at once, preventing attention to the primary objective.

41.12 Fixed Attempt Logic

The lesson advances or ends solely because a fixed number of attempts has occurred, regardless of evidence.

42. Lesson Authoring Record

Every core lesson SHOULD maintain an authoring record containing:

- lesson identifier and version;
- primary and secondary objectives;
- taxonomy domains;
- prerequisites and evidence levels;
- learner entry assumptions;
- invariant-spine mapping;
- scaffolds and fading conditions;
- likely errors and remediation;
- retrieval design;
- variation and application design;
- exit evidence and permissible claims;
- delayed-review specification;
- terminology introduced;
- accessibility considerations;
- and unresolved educational debt.

The storage format is implementation-specific. The pedagogical content is mandatory.

43. Lesson Quality Review

A lesson reviewer must evaluate:

- 1 objective-evidence alignment;
- 2 prerequisite validity;
- 3 cognitive-load control;
- 4 meaningful encounter;
- 5 directed attention;
- 6 model accuracy and timing;

- 7 learner action;
- 8 feedback usefulness;
- 9 scaffold appropriateness;
- 10 retrieval authenticity;
- 11 transfer relevance;
- 12 musical application;
- 13 exit-claim honesty;
- 14 delayed-review completeness;
- 15 terminology consistency;
- 16 tone and respect;
- 17 and conformity with non-negotiables.

A lesson may be factually correct and still fail quality review.

44. Lesson Conformance Tests

A core lesson conforms only if reviewers can answer:

- 1 What is the primary capability?
- 2 What evidence will reveal it?
- 3 Which prerequisites are required, and are they activated?
- 4 What does the learner experience before or alongside explanation?
- 5 What feature receives directed attention?
- 6 What is the minimum complete model?
- 7 What first attempt produces interpretable evidence?
- 8 What errors are likely?
- 9 How does feedback change the next attempt?
- 10 What scaffold is used, and when is it faded?
- 11 What must the learner retrieve without the full answer?
- 12 What variation tests stability or transfer?
- 13 How is the capability used musically?
- 14 What evidence supports lesson exit?
- 15 Which claims are prohibited at exit?
- 16 What happens if the learner is not ready?
- 17 When and how will delayed review occur?
- 18 How does this lesson reduce future dependence?

If these questions cannot be answered, the lesson is not ready for approval.

45. Examples of Compliant Lesson Architecture

45.1 Natural-Note Retrieval Lesson

Position: Natural-note recall provides anchors for intervals, chords, and scales. The current scope is the sixth string through the twelfth fret.

Activate: Name the open sixth string and explain the musical alphabet.

Encounter and Attend: Observe and play E, F, G, A, B, C, D, and E while noticing the half-step relationships between E-F and B-C.

Model: Demonstrate how open-string identity, fret distance, and the alphabet combine.

Attempt: Locate three named notes with a visible alphabet reference.

Interpret and Adjust: Distinguish alphabet-order errors from fret-counting errors.

Fade: Remove the alphabet reference, then remove highlighted fret markers.

Retrieve: Locate named notes in mixed order without a full diagram.

Vary: Begin from different notes and ask for adjacent notes or duplicate locations.

Apply: Use note anchors to construct a named interval or find a chord root.

Exit: Record independent same-session recall within the defined scope.

Return: Retrieve the same notes after delay, mixed with fifth-string notes.

45.2 Major-Triad Construction Lesson

Position: Major triads explain the internal structure of familiar major chords.

Activate: Recall root, major third, and perfect fifth.

Encounter: Hear and play a root alone, root plus fifth, and complete major triad.

Attend: Notice the change created by adding the major third.

Model: Derive root, third, and fifth from a named root and map them to one string set.

Attempt: Construct a second major triad with partial interval cues.

Interpret: Separate note-spelling error, interval error, and fretboard-location error.

Adjust: Rebuild the interval formula or narrow the string set.

Fade: Remove interval labels and root highlighting.

Retrieve: Construct triads from new roots.

Vary: Change root and string set.

Apply: Arpeggiate the chord over a backing harmony and identify each chord tone.

Exit: Record independent construction in familiar conditions, not comprehensive mastery.

Return: Reconstruct after delay and compare with a minor triad in a later lesson.

45.3 Eighth-Note Subdivision Lesson

Position: Subdivision determines where notes occur between beats.

Activate: Maintain and count a quarter-note pulse.

Encounter: Clap quarter notes, then hear and clap two equal attacks per beat.

Attend: Identify which attacks align with clicks and which fall between them.

Model: Demonstrate counting and physical motion at a slow tempo.

Attempt: Clap one measure with full counting support.

Interpret: Identify whether errors arise from pulse loss or unequal subdivision.

Adjust: Reduce tempo, return to spoken count, or isolate one beat.

Fade: Remove visual marks, then reduce spoken support.

Retrieve: Perform a short eighth-note and rest pattern from notation or verbal instruction.

Vary: Change attack order while preserving subdivision.

Apply: Play a one-note riff with the pattern.

Exit: Demonstrate stable same-session synchronization.

Return: Perform a new pattern after delay with less frequent click support.

46. Examples of Noncompliant Lesson Architecture

46.1 Definition-Only Lesson

A page defines scale degrees, shows a chart, and marks the lesson complete. It lacks encounter, learner action, retrieval, application, evidence, and review.

46.2 Diagram-Dependency Lesson

The learner repeatedly copies a highlighted scale pattern while every answer remains visible. The lesson cannot support independent navigation or retrieval claims.

46.3 Speed Challenge Lesson

The learner is told to increase tempo after every successful attempt without evaluating rhythm, tension, tone, or stability. The activity rewards speed-first practice.

46.4 Musical Example Without Learner Use

A riff containing chord tones is played, but the learner neither identifies nor chooses those tones. Context is present, but application is absent.

46.5 Fixed Completion Lesson

The lesson advances after ten questions regardless of guessing, prompting, or error pattern. Attempt count replaces evidence.

47. Constitutional Non-Negotiables

- 1 Every core lesson MUST identify a primary objective and evidence statement.
- 2 Every core lesson MUST activate required prerequisites or respond to their absence.
- 3 Every core lesson MUST create meaningful encounter with the target.
- 4 Every core lesson MUST require active learner response.
- 5 Every core lesson MUST provide feedback capable of changing the next attempt.
- 6 Every error repeated in practice MUST have an available recovery path.
- 7 Scaffolding MUST be temporary unless identified as a permanent musical tool.
- 8 Core content MUST be retrieved without the complete answer visible.
- 9 Core lessons MUST include variation or identify a direct transfer dependency.
- 10 Core concepts MUST connect to musical application.
- 11 Lesson completion MUST remain distinct from capability status.
- 12 Exit claims MUST NOT exceed evidence.
- 13 Delayed review MUST be defined for core content.
- 14 Lessons MUST NOT introduce multiple uncontrolled novel burdens.

- 15 Explanations MUST NOT replace experience and action.
- 16 Activities MUST NOT replace objectives.
- 17 Repetition counts MUST NOT replace quality criteria.
- 18 Tempo increases MUST follow stability.
- 19 Garden language MUST remain subordinate to technical clarity.
- 20 Lessons MUST develop increasing learner independence.

48. Tradeoffs and Resolution Rules

48.1 Consistency Versus Variety

The invariant spine creates consistency, while varied sequences preserve suitability for different domains. Authors may vary presentation but may not omit functions.

48.2 Guidance Versus Discovery

Discovery can deepen attention but may waste effort when prerequisites or boundaries are unclear. Lessons should permit meaningful noticing without requiring learners to reinvent established theory.

48.3 Explanation Versus Action

Explanation can prevent confusion, while action produces evidence. The resolution is the minimum complete model followed by learner attempt and later elaboration.

48.4 Support Versus Independence

Support enables initial success but may create dependency. Every scaffold must have a fading plan and evidence collected after reduction.

48.5 Lesson Momentum Versus Remediation

Continuing may preserve motivation, but unresolved prerequisites produce compounding failure. Required prerequisites must be remediated or the lesson must be reclassified as exposure.

48.6 Authenticity Versus Diagnosability

Musical context provides meaning, while isolated tasks clarify cause. Lessons should decompose for diagnosis and reintegrate for application.

48.7 Concision Versus Completeness

Lessons should avoid unnecessary explanation but cannot omit essential relationships, error guidance, retrieval, or review. Completeness is functional, not encyclopedic.

49. Assumptions Made

This chapter assumes:

- 1 lessons may be delivered through multiple media and interaction forms;
- 2 lesson duration will vary;
- 3 learners may enter with uneven prerequisites;

- 4 FretGarden can represent supported and independent attempts distinctly;
- 5 delayed review can occur outside the original lesson container;
- 6 different domains require different sequences while preserving the invariant spine;
- 7 some applications will be deferred across a short, explicit sequence;
- 8 authors can document scaffolds, errors, evidence, and review obligations;
- 9 lesson completion and capability status can remain conceptually separate; and
- 10 later chapters will define exercise, assessment, progression, and review systems in greater detail.

50. Unresolved Questions

The following matters remain unresolved:

- 1 whether learner-facing lessons should visibly identify all architectural phases;
- 2 standard limits for the number of secondary objectives;
- 3 default rules for lesson splitting;
- 4 domain-specific minimum application requirements;
- 5 standard timing of within-lesson retrieval;
- 6 the degree to which learners may skip modeled instruction after diagnostic success;
- 7 requirements for offline or instrument-free lesson variants;
- 8 accessibility-specific scaffold alternatives;
- 9 criteria for automated classification of error causes;
- 10 and how lesson architecture should adapt when live instructor feedback is available.

51. Topics Intentionally Deferred

Deferred topics include:

- 1 exercise architecture;
- 2 assessment construction;
- 3 mastery thresholds;
- 4 prerequisite graphs;
- 5 progression rules;
- 6 review scheduling;
- 7 adaptive sequencing;
- 8 remediation depth;
- 9 feedback taxonomy and wording;
- 10 domain-specific practice prescriptions;
- 11 creative-work rubrics;
- 12 accessibility accommodations;

- 13 placement and lesson bypass;
- 14 lesson analytics;
- 15 authoring workflow and approval governance;
- 16 and implementation-specific content schemas.

52. Required Future Cross-References

52.1 Exercise Design Chapter

Must reference the invariant spine, attempt design, feedback architecture, error recovery, retrieval, variation, and practice requirements.

52.2 Assessment Chapter

Must reference exit evidence, claim limitations, prompt fading, retrieval authenticity, and delayed review.

52.3 Mastery and Progression Chapter

Must reference activation failure, exit status, readiness claims, and the distinction between completion and capability.

52.4 Review and Retention Chapter

Must reference the Return phase and delayed-review architecture.

52.5 Feedback and Remediation Chapters

Must reference Interpret, Adjust, the error-recovery loop, dimensional feedback, and remediation routing.

52.6 Domain-Specific Pedagogy Chapters

Must adapt the invariant spine without weakening its obligations.

53. Chapter Conformance Statement

A FretGarden lesson conforms to this chapter only when it translates a defined objective into a coherent sequence of experience, attention, model, action, feedback, correction, scaffold reduction, retrieval, variation, application, evidence-based exit, and delayed return.

The visible form may vary. The educational obligations may not.

A lesson is not complete merely because its content has been presented or its activities have been finished. It is complete as an instructional design only when it creates a defensible path from entry condition to evidence, defines what the evidence permits FretGarden to claim, and establishes how the capability will be revisited after time has passed.